

# Architecture and Preprocessing Choices under Dataset Shift in Single-Channel EEG Sleep Staging: A Controlled Evaluation on Sleep-EDF and SHHS1

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## Abstract

**Background.** Cross-cohort performance losses in automatic sleep staging can be attributed incorrectly when architecture, preprocessing and evaluation protocol change simultaneously. We asked which development axis—architecture replacement, signal preprocessing or targeted adaptation—was best supported before deploying a single-channel EEG model to a cohort with a different derivation and population.

**Methods.** Six active configurations were evaluated on the Sleep Cassette subset of Sleep-EDF Expanded (78 subjects, 153 recordings and 195,469 scored 30 s epochs; Fpz–Cz) using one 10-fold subject-wise partition and one validation-based model-selection rule. Pre-specified contrasts changed the sequence model, epoch encoder or signal handling while holding the remaining components fixed. Paired subject-cluster bootstrap intervals (10,000 resamples) and two-sided Wilcoxon signed-rank tests with Holm correction quantified uncertainty. Generalisation was tested by locked zero-shot inference on 180 SHHS Visit-1 participants (169,012 scored epochs; C4–A1), without weight updates on SHHS.

**Results.** The ResNet-1D–TCN fixed-scale configuration (E3) attained the highest out-of-fold macro-F1 on Sleep-EDF (0.7904). Relative to per-record z-scoring (E6), the difference was 0.0213 (CI<sub>95%</sub> [0.0122, 0.0307]; Holm  $p = 0.0012$ ); a second fixed seed preserved the direction but not Holm significance. The isolated architectural contrasts were small and non-significant. In a controlled Tesla V100 benchmark, the ResNet-1D–TCN pipeline had 3.76× lower forward-pass latency than the 15-branch CNN–BiLSTM control, although it used 4.37× as many parameters and 28.4% more peak GPU memory. Observed training-and-validation wall-clock time was 10.2–12.2× lower for the ResNet-1D–TCN configurations. On SHHS1, E3 exceeded the control by 0.0412 subject-mean macro-F1 (CI<sub>95%</sub> [0.0314, 0.0512]; Holm  $p = 2.65 \times 10^{-13}$ ) and E6 by 0.0274. Its pooled macro-F1 nevertheless fell to 0.6099. Its overall advantage did not extend to N3: E0 and E3 assigned 72.3% and 73.1% of true N3 epochs to N2 and achieved near-identical N3 recall (0.2610 and 0.2582). Thus, N3 under-detection was a shared transfer failure across the two evaluated pipelines rather than an E3-specific defect. E6 did not resolve it (N3 recall 0.2005 overall and 0.0721 near transitions, versus 0.0733 for E3).

**Conclusions.** The architectural substitutions offered an operational advantage but did not establish a stable predictive advantage. Preprocessing was the stronger development axis in this experiment, while the dominant transfer errors identify N2–N3 and N2–REM as priorities for labelled target-domain adaptation. Record-wise normalisation alone was not a sufficient remedy.

**Keywords:** sleep staging; single-channel EEG; ResNet-1D; temporal convolutional network; Sleep-EDF Expanded; Sleep Heart Health Study; domain shift; zero-shot evaluation; paired inference.

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## 1. Introduction

Sleep staging assigns one of five labels—W, N1, N2, N3 and REM—to every 30 s epoch of a polysomnographic recording. Manual scoring is slow, requires trained expertise, and retains irreducible disagreement between scorers; in the inter-scorer reliability programme of the American Academy of Sleep Medicine, N1 shows the lowest agreement of all stages [1]. An automated system therefore has to be judged not by overall accuracy alone, which is dominated by the most frequent stages, but by class-balanced metrics, by stability across individuals, and by its behaviour when the recording device, montage or population changes.

Progress in this field is difficult to read from published tables. Studies differ simultaneously in dataset version, subject partitioning, wake-trimming policy, filtering, normalisation, sequence length and model-selection rule, so a reported improvement cannot be attributed to the architecture that the paper nominally proposes. This creates a concrete development problem: before collecting target-domain labels, a practitioner cannot tell whether a new encoder, a new sequence model or a change in signal handling is the more defensible use of effort. A comparison in which one component changes at a time is needed to separate these decisions.

We take ZleepAnlystNet [2], a two-stage single-channel model that couples a bank of class-specific convolutional encoders to a BiLSTM, as an internal control, and we substitute two components in isolation: a temporal convolutional network (TCN) as a parallelisable sequence model [3], and a one-dimensional residual network (ResNet-1D) as an epoch encoder [4]. Neither substitution is assumed beneficial. We additionally treat the signal-preprocessing scheme as an experimental factor rather than an implementation detail, because filtering and record-wise normalisation are common responses to acquisition differences and can change the information available to a downstream classifier.

The study is organised around three deployment-oriented questions. (i) Under a shared subject-wise protocol, do the TCN and ResNet-1D substitutions provide incremental benefits large and stable enough to justify changing the architecture? (ii) Which evaluated pipeline and preprocessing choices retain their benefit under zero-shot transfer to an independent cohort with a different EEG derivation and age distribution? (iii) When transfer performance falls, which class transitions account for the loss and do they point to a generic preprocessing remedy or to targeted adaptation?

Our contributions are the following.

- (1) A leakage-controlled 10-fold subject-wise protocol that estimates the incremental effect of sequence-model, encoder and preprocessing changes under shared partitions and model-selection rules.
- (2) A locked zero-shot evaluation on 180 SHHS Visit-1 participants, showing that the ranking and magnitude of design choices can differ between internal validation and cross-cohort use.
- (3) A class- and confusion-channel analysis showing that under-detection of N3 as N2 recurs in both the 15-CNN–BiLSTM control and ResNet-1D–TCN, while over-detection of REM from N2 is a second E3 priority. The accompanying E6 sensitivity analysis shows that per-record normalisation alone does not resolve the cross-model N3 failure.
- (4) A quantified operational trade-off: the implemented ResNet-1D–TCN pipeline reduces measured forward latency and training-and-validation wall-clock time relative to the 15-branch CNN–BiLSTM control, while increasing parameter count and peak memory and without establishing predictive superiority.

Throughout, we distinguish pre-specified comparisons, secondary evidence and post-hoc contrasts, and we confine claims to the evidence class that supports them.

## 2. Related work

DeepSleepNet combined a dual-branch CNN with a BiLSTM to capture epoch morphology and temporal dependence [5]. AttnSleep introduced attention over multi-resolution features [6], and SleepTransformer applied self-attention with explicit uncertainty quantification [7]. Together these establish that sequential context matters substantially. They also illustrate the comparability problem: differences in dataset version, subject split and preprocessing between these works are large enough that cross-table comparison cannot substitute for an ablation carried out inside one protocol, a point emphasised in recent reviews of the field [8].

Cross-dataset sleep staging is already an active methodological problem. For example, He et al. [9] trained a domain-adaptation network across Sleep-EDF, SHHS1 and the PhysioNet Challenge data, while Van Der

Donckt et al. [10] showed that increasingly complex deep architectures need not outperform a carefully designed conventional pipeline under a matched protocol. Our study does not propose another adaptation algorithm. It addresses the decision immediately before adaptation: under zero-shot transfer, which existing pipeline choices remain useful, and which class boundaries should receive the limited target-domain labels?

Sleep-EDF Expanded is the standard public benchmark for single-channel EEG [11, 12]. The Sleep Heart Health Study (SHHS) is a large multi-centre cohort assembled to study sleep-disordered breathing and cardiovascular risk [13], distributed through the National Sleep Research Resource (NSRR) [14]. The two differ in age, comorbidity, recording hardware, montage and label distribution. This makes SHHS a considerably more realistic transfer test than a held-out split of the same corpus, and it is the reason we treat the transfer result—not the in-domain ranking—as the principal evidence about deployability.

Two threads of the literature bear directly on our findings. First, the sleep slow oscillation has a pronounced anterior amplitude maximum and propagates as a travelling wave [15], so the microvolt amplitude of slow-wave activity is derivation-dependent; and slow-wave amplitude declines markedly with age [16]. Both scoring manuals [17, 18] define deep sleep partly through an absolute amplitude criterion, which means an amplitude criterion learned on one derivation and one age band need not transfer. Second, classifier outputs trained under one label prior are miscalibrated under another, and the correction is well characterised [19, 20]. These observations motivate the class-wise transfer analysis in Section 4.6; because cohort and derivation change together, they are treated as candidate explanations rather than mechanisms identified by this design.

### 3. Materials and methods

#### 3.1. Data and labels

The Sleep Cassette subset of Sleep-EDF Expanded comprises 153 recordings from 78 subjects; we use the Fpz–Cz derivation at 100 Hz with 30 s epochs (3,000 samples per epoch). Annotations are mapped to W, N1, N2, N3 and REM; Movement and Unknown epochs retain their temporal position but are excluded from the loss and from all metrics. After windowing, 195,767 epochs remain, of which 195,469 are scored and 298 are Movement/Unknown.

An evaluation window is applied identically to both cohorts. It is anchored on the first and last true sleep epoch (any of N1, N2, N3, REM) and extended by 30 min of wake on each side, i.e. 60 epochs per side. This policy is stated explicitly because it strongly affects the wake fraction and hence every aggregate metric, and because it is one of the principal reasons published numbers are not directly comparable. It is a benchmark window, not a deployable inference rule, since its anchors use ground-truth labels.

The SHHS Visit-1 sample was locked before any prediction was inspected and comprises 180 test participants, with a further 5 adaptation and 15 validation participants that were never used to update weights. The EEG channel is C4–A1 at 125 Hz, resampled to 100 Hz with a polyphase anti-aliasing filter (up 4, down 5, Kaiser window  $\beta = 5.0$ ) applied to the continuous signal before epoching. SHHS labels are mapped  $0 \rightarrow W$ ,  $1 \rightarrow N1$ ,  $2 \rightarrow N2$ ,  $\{3, 4\} \rightarrow N3$ ,  $5 \rightarrow REM$ ,  $\{6, 9\} \rightarrow \text{excluded}$ . Table 1 reports both label distributions.

The prior shift between cohorts is substantial: total variation distance 0.187,  $D_{KL}(\text{EDF} \parallel \text{SHHS}) = 0.101$  nats. SHHS1 contains proportionally  $2.02\times$  as much N3 and  $0.38\times$  as much N1. We report the per-class ratios in Table 1 because Section 4.6 uses their *direction* as evidence.

#### 3.2. Preprocessing conditions

All conditions operate on the continuous calibrated signal in microvolts before epoching. Four active signal conditions enter the performance analysis, and one audit-only condition (E5) records a planned clipping ablation:

- **Raw** (E0–E2): no filtering, no clipping, no rescaling; native  $\mu V$  retained.
- **Band-pass only** (E4): zero-phase Butterworth band-pass, order 4, 0.5–30 Hz, implemented as second-order sections with forward–backward filtering.
- **Band-pass plus clipping** (E5; audit only): the E4 signal followed by a hard clip at  $\pm 800 \mu V$ . Across all 153 Sleep-EDF recordings, E5 and E4 were bitwise identical and the measured clipped-sample fraction was zero. Fold 0 was retained as an audit artifact; the remaining folds and all performance tests were

**Table 1.** Scored-label distribution in the two evaluation samples.

| Stage               | Sleep-EDF      |             | SHHS1 test     |             |              |
|---------------------|----------------|-------------|----------------|-------------|--------------|
|                     | Epochs         | Share       | Epochs         | Share       | Ratio to EDF |
| W                   | 65,941         | 33.73%      | 36,983         | 21.88%      | 0.65         |
| N1                  | 21,522         | 11.01%      | 7,002          | 4.14%       | 0.38         |
| N2                  | 69,132         | 35.37%      | 76,287         | 45.14%      | 1.28         |
| N3                  | 13,039         | 6.67%       | 22,806         | 13.49%      | 2.02         |
| REM                 | 25,835         | 13.22%      | 25,934         | 15.34%      | 1.16         |
| <b>Total scored</b> | <b>195,469</b> | <b>100%</b> | <b>169,012</b> | <b>100%</b> |              |

**Table 2.** Experimental configuration codes. The identifiers denote signal–model combinations, not a performance ranking or a sequence of model versions.

| ID | Signal condition               | Epoch encoder | Sequence model and role                |
|----|--------------------------------|---------------|----------------------------------------|
| E0 | Raw                            | 15-branch CNN | BiLSTM; internal control               |
| E1 | Raw                            | 15-branch CNN | TCN; isolates the sequence model       |
| E2 | Raw                            | ResNet-1D     | TCN; isolates the epoch encoder        |
| E3 | Band-pass + clip + fixed scale | ResNet-1D     | TCN; fixed-scale configuration         |
| E4 | Band-pass only                 | ResNet-1D     | TCN; effect of filtering alone         |
| E5 | Band-pass + clip               | ResNet-1D     | TCN; audit only, identical input to E4 |
| E6 | Band-pass + clip + z-score     | ResNet-1D     | TCN; alternative rescaling             |

cancelled before scores were inspected because E5 did not define an independent input condition.

- **Fixed-scale amplitude handling** (E3): the same band-pass, followed by a hard clip at  $\pm 800 \mu\text{V}$  and division by a constant factor of 100. The clip threshold is an absolute constant identical for every recording, so no record-level or cohort-level statistic is estimated and cross-subject leakage is not introduced; relative amplitude between recordings is preserved.
- **Per-record z-score** (E6): the same band-pass and clip, followed by subtraction of the mean and division by the standard deviation computed over the whole filtered, clipped recording. This is label-independent but removes absolute amplitude differences between recordings.

No notch filter is applied: 50 Hz lies outside the 0.5–30 Hz pass-band and at the Nyquist frequency of the 100 Hz signal. The contrast E3–E6 therefore evaluates a practical preprocessing choice: preserving calibrated cross-recording amplitude relationships or applying record-wise normalisation. Because clipping never activated in E5, E3 and E4 differ in these data by the fixed division by 100; this identity is used only to interpret the observed contrast and does not establish equivalence or a hardware-independent scaling rule.

### 3.3. Model configurations

Table 2 defines seven configuration codes. Six active configurations (E0–E4 and E6) enter performance analysis; E5 is retained solely to document why the clipping-only contrast was removed. The E labels are compact configuration identifiers used consistently across artifacts and analyses.

**15-branch CNN encoder.** The control encoder instantiates 15 independent small CNNs: three context groups—C (current epoch), P (previous), N (next)—crossed with five class-specific detectors (W, N1, N2, N3, REM). Each branch has 2,495 parameters and comprises two convolutional blocks, Conv1d( $1 \rightarrow 10, k=55$ ) and Conv1d( $10 \rightarrow 5, k=25$ ), each followed by ReLU, batch normalisation and  $16 \times$  max-pooling, then a  $60 \rightarrow 10 \rightarrow 5$  head. The concatenated output is a 75-dimensional vector per epoch ( $15 \text{ branches} \times 5 \text{ softmax outputs}$ ), ordered C, P, N. Context shifts are clamped at recording boundaries.

**ResNet-1D encoder.** A stem Conv1d( $1 \rightarrow 32, k=50, s=5$ ) with batch normalisation, ReLU and max-pooling is followed by three residual blocks ( $32 \rightarrow 64, s=1$ ;  $64 \rightarrow 128, s=2$ ;  $128 \rightarrow 128, s=2$ ), each containing two  $k=7$  convolutions with a projection shortcut where shape changes. Global average pooling yields a 128-dimensional epoch embedding; total temporal downsampling is  $40\times$ . Note that the ResNet-1D encoder consumes *only the current epoch*: it has no C/P/N construction.

**Sequence models.** The TCN projects the encoder output to 128 dimensions and applies six residual blocks of two  $k=3$  convolutions each, with dilations 1, 2, 4, 8, 16, 32, LayerNorm, GELU and dropout 0.2. Padding is symmetric, so the network is *non-causal*, and the receptive field is  $1 + 2(k-1) \sum_i d_i = 1 + 4 \times 63 = 253$  epochs, i.e. approximately  $\pm 126$  epochs or  $\pm 63$  min around the epoch being classified. The BiLSTM control is a single bidirectional layer with 128 hidden units per direction. In both cases the input is the *entire overnight recording* as one sequence (typically 900–1,200 epochs), not a short window; padding is masked at every layer. The receptive-field figure is reported to make the temporal support of the TCN explicit.

### 3.4. Two-stage training and subject-wise cross-validation

The 78 subjects are partitioned into 10 subject-disjoint folds. In each evaluation round one fold is the test fold, the next fold (modulo 10) is the validation fold and the remainder are training folds. Both nights of a subject always share the same role. The same partition is used for every configuration, and the test fold never participates in model selection; the training API accepts only training and validation loaders, so test data cannot enter fitting.

Training proceeds in two stages with a *fully frozen* encoder. Stage 1 fits the epoch encoder at epoch level: for E0/E1, the 15 CNN branches are trained independently as five-class problems under class-specific binary weights  $w_c = 0.5N/n_c$  and  $w_{j \neq c} = 0.5N/(N - n_c)$  computed from training subjects only; for E2–E6, the ResNet-1D and a linear head are trained on single epochs under balanced weights  $w_c = N/(5n_c)$ , again from training subjects only. The best checkpoint is then verified by SHA-256, switched to evaluation mode, and used under `torch.no_grad()` to extract and cache features. Stage 2 fits the sequence model on those cached features; because the inputs are arrays, the encoder is not in the computation graph and the freeze is absolute.

Optimisation settings are: 15-branch CNN, Adam, lr  $10^{-3}$ , batch 64 epochs, patience 10; ResNet-1D, AdamW, lr  $10^{-3}$ , weight decay  $10^{-4}$ , batch 64, max 40 epochs, patience 8; BiLSTM, Adam, lr  $10^{-2}$ , batch 4 recordings, patience 10; TCN, Adam, lr  $5 \times 10^{-4}$ , batch 8 recordings, max 300 epochs, patience 30. Gradient-norm clipping at 1.0 is applied to sequence models. There is no learning-rate schedule, no label smoothing and no data augmentation. Sequence models use unweighted masked cross-entropy, so their optimisation target reflects the empirical Sleep-EDF class distribution; output probabilities were not separately calibrated. Model selection uses validation macro-F1 (validation weighted loss for the CNN branches). E1 reuses the E0 encoder checkpoints unchanged, so E1 – E0 isolates the sequence-model substitution exactly.

### 3.5. Test locking, statistical analysis and effect sizes

All six configurations completed validation-based selection before the test set was opened, once. In Sleep-EDF, pooled macro-F1 is the descriptive headline metric, but the unit of statistical inference is the subject. Paired subject-cluster bootstrap with 10,000 resamples estimates confidence intervals for the difference in pooled macro-F1, while a two-sided Wilcoxon signed-rank test [21] assesses the subject-level distribution of per-subject differences [22]. Holm correction [23] is applied over the four pre-specified comparisons E1 – E0, E2 – E1, E3 – E2 and E3 – E6. These two procedures target different estimands—a pooled, epoch-weighted quantity and a subject-averaged rank quantity—and may therefore disagree; where they do, we say so explicitly rather than reporting only the favourable one. E3 – E0 is labelled post-hoc throughout.

To make the direction of the paired effect transparent, we additionally report a sign-dominance statistic for every comparison,

$$\Delta_{\text{sign}} = \frac{W - L}{W + L}, \quad (1)$$

where  $W$  and  $L$  are the numbers of subjects improved and degraded. The statistic equals  $2\hat{p}_{\text{win}} - 1$ , lies in  $[-1, 1]$ , and is accompanied by an exact two-sided binomial sign test. It deliberately ignores effect

magnitude and is therefore reported as an unadjusted descriptive robustness measure, not as a member of the pre-specified inferential family.

After the primary results had been observed, the whole campaign was repeated with a second fixed training seed on the same partition. The two seeds are analysed separately;  $p$ -values are never pooled, and two fixed seeds are not treated as a sample from the distribution over initialisations. This repeat is reported as a post-protocol sensitivity analysis.

For SHHS, each configuration uses all ten Sleep-EDF outer-fold checkpoints. Softmax probabilities are averaged in float64 across folds before the final argmax for each epoch. The pre-locked primary criterion is subject-mean macro-F1, with E3 – E0 and E3 – E6 as the two locked comparisons under the same bootstrap and Wilcoxon–Holm treatment. The E1/E2 component analysis is secondary evidence on an already-opened sample, and E3 – E2 is explicitly post-hoc.

### 3.6. Operational evaluation

Inference timing was measured on identical hardware (Tesla V100) for all configurations and excludes I/O, preprocessing and training. Each measurement used a batch of one synthetic 100-epoch sequence, 20 warm-up passes and three rounds of 100 timed forward passes. Configuration order was shuffled in each round; we report the median over all 300 passes. This benchmark measures the implemented full forward pipeline for one fixed tensor shape, not overnight wall-clock inference or algorithmic complexity.

## 4. Results

### 4.1. Out-of-fold performance on Sleep-EDF

E3 is descriptively highest on all four metrics (Table 3). A pooled ranking is not by itself statistical evidence, and Table 4 shows how little of this ordering survives inference.

Only E3 – E6 is consistent across all three lenses: a fully positive bootstrap interval, Holm significance, and a sign test that agrees. The remaining pre-specified contrasts have intervals containing zero and do not reach significance.

E3 – E2 deserves particular attention because it is the clearest case of estimand disagreement in the study. Its pooled bootstrap interval marginally excludes zero ( $[0.000305, 0.014520]$ ), yet only 37 of 78 subjects improve,  $\Delta_{\text{sign}} = -0.051$ , and both the Wilcoxon and sign tests are far from significance ( $p_{\text{sign}} = 0.734$ ). Pooled macro-F1 is epoch-weighted and is therefore dominated by subjects contributing more scored epochs; the subject-level statistics weight every subject equally. When these disagree, the honest reading is that the effect is carried by a subset of high-epoch-count subjects and is not a general property of the population. We do not claim E3 improves on E2 in-domain.

For the post-hoc E3 – E0 contrast, the added macro-F1 of the complete pipeline is 0.0150 with a fully positive interval; because it changes architecture and preprocessing simultaneously, it supports the pipeline as a whole and cannot attribute credit to any component.



**Table 3.** Out-of-fold test performance on Sleep-EDF (seed 42, pooled over all folds).

| Configuration | macro-F1      | Accuracy      | $\kappa$      | F1 (N1)       |
|---------------|---------------|---------------|---------------|---------------|
| E0            | 0.7754        | 0.8262        | 0.7614        | 0.5009        |
| E1            | 0.7802        | 0.8315        | 0.7683        | 0.5115        |
| E2            | 0.7835        | 0.8341        | 0.7716        | 0.5180        |
| E3            | <b>0.7904</b> | <b>0.8363</b> | <b>0.7754</b> | <b>0.5276</b> |
| E4            | 0.7891        | 0.8360        | 0.7746        | 0.5270        |
| E6            | 0.7691        | 0.8230        | 0.7573        | 0.5124        |

**Table 4.** Pre-specified paired comparisons on Sleep-EDF macro-F1 ( $n = 78$  subjects). Holm correction is over the four pre-specified rows; sign-test statistics are unadjusted supplementary robustness checks. E3 – E0 is post-hoc and is excluded from the Holm family.

| Comparison                | $\Delta$ macro-F1 | CI <sub>95%</sub>           | $p_{\text{Holm}}$ | W/T/L   | $\Delta_{\text{sign}}$ | $p_{\text{sign}}$ |
|---------------------------|-------------------|-----------------------------|-------------------|---------|------------------------|-------------------|
| E1 – E0                   | 0.004811          | [−0.000915, 0.010193]       | 0.1022            | 51/0/27 | +0.308                 | 0.0088            |
| E2 – E1                   | 0.003251          | [−0.002370, 0.008808]       | 0.1036            | 44/0/34 | +0.128                 | 0.3082            |
| E3 – E2                   | 0.006962          | [0.000305, 0.014520]        | 0.8989            | 37/0/41 | −0.051                 | 0.7343            |
| E3 – E6                   | <b>0.021319</b>   | <b>[0.012179, 0.030698]</b> | <b>0.0012</b>     | 49/0/29 | <b>+0.256</b>          | <b>0.0308</b>     |
| <i>E3 – E0 (post-hoc)</i> | 0.015024          | [0.005746, 0.025871]        | —                 | 49/0/29 | +0.256                 | 0.0308            |

#### 4.2. Per-class error structure on Sleep-EDF

Table 5 breaks performance down by stage. N1 is by far E3’s hardest class, which is consistent with it also being the stage of lowest human inter-scorer agreement [1]. The largest advantage of E3 over E6 is on N3 (+0.0628), and the gain is driven mainly by precision (0.807 vs 0.716). This is the first indication that the choice between fixed scaling and record-wise normalisation changes the N3 decision, which motivates examining N3 explicitly in the transfer analysis in Section 4.6.

A useful summary of in-domain marginal class emission is the ratio of predicted to true epochs per class. For E3 on Sleep-EDF these ratios are W 0.993, N1 1.051, N2 0.984, N3 1.003, REM 1.017: the model closely reproduces the marginal label distribution. This near-match is the descriptive baseline against which the SHHS behaviour is read; it is not a probability-calibration assessment.

In the pooled confusion matrix (Table 6), the dominant error channels are the reciprocal N2↔N1 confusions (5,496 and 4,867) and W→N1 (3,888), i.e. the in-domain error budget is concentrated on N1 boundaries—the expected behaviour for a well-fitted single-channel model.

**Table 5.** Per-class metrics on Sleep-EDF: E3 versus E6 (pooled out-of-fold predictions, seed 42). Descriptive statistics on the same test predictions; not a new test family.

| Stage | E3 Prec. | E3 Rec. | E3 F1  | E6 Prec. | E6 Rec. | E6 F1  | $\Delta F1$    |
|-------|----------|---------|--------|----------|---------|--------|----------------|
| W     | 0.9338   | 0.9271  | 0.9304 | 0.9391   | 0.9169  | 0.9279 | +0.0026        |
| N1    | 0.5149   | 0.5410  | 0.5276 | 0.5032   | 0.5219  | 0.5124 | +0.0152        |
| N2    | 0.8583   | 0.8447  | 0.8514 | 0.8401   | 0.8368  | 0.8385 | +0.0130        |
| N3    | 0.8072   | 0.8097  | 0.8085 | 0.7160   | 0.7779  | 0.7457 | <b>+0.0628</b> |
| REM   | 0.8274   | 0.8413  | 0.8343 | 0.8227   | 0.8197  | 0.8212 | +0.0131        |

**Table 6.** Pooled confusion matrix of E3 on Sleep-EDF (rows: reference; columns: prediction).

| Reference \ Predicted | W      | N1     | N2     | N3     | REM    |
|-----------------------|--------|--------|--------|--------|--------|
| W                     | 61,133 | 3,888  | 416    | 56     | 448    |
| N1                    | 3,283  | 11,643 | 4,867  | 105    | 1,624  |
| N2                    | 539    | 5,496  | 58,394 | 2,271  | 2,432  |
| N3                    | 30     | 45     | 2,377  | 10,558 | 29     |
| REM                   | 483    | 1,542  | 1,984  | 90     | 21,736 |

### 4.3. Sensitivity to the training seed

Table 7 lists the per-configuration figures. All four pre-specified differences keep a positive sign in both seeds, and E3 – E6 is the only contrast whose bootstrap interval is fully positive in both runs. However, its Holm-adjusted significance appears in seed 42 only, and its magnitude halves ( $0.0213 \rightarrow 0.0102$ ). The defensible conclusion is that the *direction* of the amplitude-handling contrast is stable while its *inferential strength* is not. E6 shows the largest absolute seed-to-seed change (+0.0089), but two fixed seeds cannot identify its cause, characterise the distribution over initialisations or constitute an independent replication.



**Table 7.** Out-of-fold test macro-F1 for two fixed seeds. Seed 42 is the primary campaign; seed 123 is a post-protocol repeat on the same partition.

| Configuration | Seed 42 | Seed 123 | $\Delta$ |
|---------------|---------|----------|----------|
| E0            | 0.7754  | 0.7755   | +0.0000  |
| E1            | 0.7802  | 0.7776   | -0.0026  |
| E2            | 0.7835  | 0.7828   | -0.0007  |
| E3            | 0.7904  | 0.7883   | -0.0022  |
| E4            | 0.7891  | 0.7887   | -0.0004  |
| E6            | 0.7691  | 0.7780   | +0.0089  |

**Table 8.** Seed sensitivity of the four pre-specified contrasts. Each  $p$ -value is Holm-corrected *within* its own seed; no test pools the two seeds.

| Comparison | Seed 42: $\Delta$ [CI], $p_{\text{Holm}}$    | Seed 123: $\Delta$ [CI], $p_{\text{Holm}}$ |
|------------|----------------------------------------------|--------------------------------------------|
| E1 – E0    | 0.004811 [-0.000915, 0.010193], 0.1022       | 0.002188 [-0.002870, 0.007043], 0.9130     |
| E2 – E1    | 0.003251 [-0.002370, 0.008808], 0.1036       | 0.005143 [-0.000571, 0.011292], 0.2400     |
| E3 – E2    | 0.006962 [0.000305, 0.014520], 0.8989        | 0.005478 [-0.002611, 0.015796], 0.9130     |
| E3 – E6    | 0.021319 [0.012179, 0.030698], <b>0.0012</b> | 0.010249 [0.002435, 0.017989], 0.1313      |

#### 4.4. Complexity, latency and training cost

As Table 9 shows, the ResNet-1D–TCN family has  $3.76\times$  lower measured latency than the CNN–BiLSTM control while carrying  $4.37\times$  the parameters and 28.4% more peak allocated memory. The implementation is therefore faster but not smaller. E1 differs from E0 only in the sequence model and improves latency by just  $1.07\times$ ; the larger E2–E0 difference is consequently associated primarily with replacing 15 independently executed CNN branches by one ResNet-1D encoder, together with the full-pipeline implementation. The benchmark does not isolate kernel-launch overhead or predict latency for other sequence lengths and hardware.

For the complete training-and-validation workflow (Table 10), E2 finished  $12.2\times$  faster than E0 and E3  $10.2\times$  faster. E1’s 14 min total is not an end-to-end comparison because it reuses the E0 encoder checkpoints and pays only the sequence-model stage; it nevertheless shows that the 15-branch encoder training dominates the control workflow. These are observed wall-clock costs under one scheduling regime, early-stopping policy and GPU, not hardware-independent complexity claims.

**Table 9.** Full-pipeline forward benchmark on a Tesla V100 (fold 0, seed 42; batch 1, sequence length 100). Throughput is in epochs  $s^{-1}$  and peak allocated memory in MiB; I/O and preprocessing are excluded.

| ID | Parameters | Median (ms) | Throughput | Peak mem. | Speed-up |
|----|------------|-------------|------------|-----------|----------|
| E0 | 248,630    | 13.4315     | 7,445      | 58.81     | 1.00×    |
| E1 | 640,950    | 12.5111     | 7,993      | 60.31     | 1.07×    |
| E2 | 1,085,578  | 3.5750      | 27,972     | 75.54     | 3.76×    |
| E3 | 1,085,578  | 3.5687      | 28,021     | 75.54     | 3.76×    |
| E6 | 1,085,578  | 3.5756      | 27,967     | 75.54     | 3.76×    |

**Table 10.** Wall-clock training and validation cost of the seed-123 campaign (10 folds, sequential, same GPU). The test set was not opened during these runs.

| Configuration                     | Total (10 folds)        | Mean per fold          |
|-----------------------------------|-------------------------|------------------------|
| E0 — 15-branch CNN + BiLSTM       | 33 h 35 min 36 s        | 3 h 21 min 34 s        |
| E1 — 15-branch CNN + TCN          | 14 min 17 s             | 1 min 26 s             |
| E2 — ResNet-1D + TCN              | 2 h 44 min 57 s         | 16 min 30 s            |
| E3 — ResNet-1D + TCN, fixed scale | 3 h 16 min 40 s         | 19 min 40 s            |
| E4 — ResNet-1D + TCN, band-pass   | 2 h 55 min 56 s         | 17 min 36 s            |
| E6 — ResNet-1D + TCN, $z$ -score  | 2 h 24 min 44 s         | 14 min 28 s            |
| <b>All six configurations</b>     | <b>45 h 12 min 10 s</b> | <b>4 h 31 min 13 s</b> |

#### 4.5. Zero-shot evaluation on SHHS1

The second-seed campaign (Table 12) preserves the direction and Holm significance of both locked SHHS contrasts, whereas the in-domain E3 – E6 comparison no longer reaches the Holm threshold under seed 123 (Table 8). This strengthens the evidence that the locked cross-cohort ordering is not unique to one training seed. It is a sensitivity analysis on the same SHHS sample, not an independent cohort replication.

Both locked comparisons (Table 13) are unambiguous, and the sign-based effect sizes are large ( $\Delta_{\text{sign}} = +0.533$  and  $+0.389$ ; exact sign tests  $p = 3.8 \times 10^{-13}$  and  $1.9 \times 10^{-7}$ ). Supporting metrics move together: relative to E0, E3 gains 0.0338 pooled macro-F1, 0.0368 accuracy and 0.0461  $\kappa$ . Within  $\pm 1$  epoch of a transition, E3 gains 0.0366 over E0 and 0.0182 over E6.

The component analysis is more equivocal. E1 – E0 remains positive but small, supporting the TCN substitution. E2 – E1 is *negative* out of domain ( $-0.0128$ , interval entirely below zero), directly contradicting the in-domain ordering: the ResNet-1D encoder does not generalise better than the 15-branch CNN encoder on unprocessed SHHS signals. The post-hoc E3 – E2 contrast is by far the largest effect in the entire study ( $+0.0475$ ,  $\Delta_{\text{sign}} = +0.633$ , 147/180 subjects improved). Since E3 and E2 share an identical architecture and differ only in a preprocessing bundle, the result makes preprocessing the stronger development axis in this experiment. Its post-hoc status prevents treating the contrast as confirmatory evidence for a particular operation.

A seed-123 extension evaluating E4 (band-pass only) provides a more specific secondary contrast. E4 reached subject-mean macro-F1 0.5732, pooled macro-F1 0.6147, accuracy 0.7064,  $\kappa = 0.5869$  and N1 F1 0.3492, exceeding same-seed E2 by 0.0323 and E3 by 0.0098. The E5 audit established that the  $\pm 800 \mu V$  clip never activated, so E4 and E3 differ in this dataset by the fixed division by 100. The extension therefore associates band-pass filtering with most of the E4–E2 gain and provides no evidence that the additional fixed scaling improves transfer. Because the SHHS sample was already open and training stochasticity is represented by only two fixed seeds, these comparisons are secondary; they are not equivalence, non-inferiority or universal scaling tests.

**Table 11.** Zero-shot performance on 180 locked SHHS1 participants (seed 42). No weights were updated on SHHS. Intervals are subject-cluster bootstrap intervals for the subject-mean macro-F1. Transition macro-F1 is restricted to the 44,744 epochs within  $\pm 1$  of a stage change (26.5% of the sample).

| ID | Subject-mean macro-F1 (CI <sub>95%</sub> ) | Pooled macro-F1 | Accuracy      | $\kappa$      | F1 (N1)       | Transition macro-F1 |
|----|--------------------------------------------|-----------------|---------------|---------------|---------------|---------------------|
| E0 | 0.5268 [0.5101, 0.5431]                    | 0.5761          | 0.6648        | 0.5339        | 0.2998        | 0.4323              |
| E1 | 0.5333                                     | 0.5822          | 0.6709        | 0.5409        | 0.3197        | 0.4352              |
| E2 | 0.5205                                     | 0.5668          | 0.6656        | 0.5327        | 0.2833        | 0.4219              |
| E3 | <b>0.5680 [0.5503, 0.5850]</b>             | <b>0.6099</b>   | <b>0.7016</b> | <b>0.5801</b> | <b>0.3451</b> | <b>0.4689</b>       |
| E6 | 0.5407 [0.5227, 0.5584]                    | 0.5732          | 0.6742        | 0.5408        | 0.3102        | 0.4507              |

**Table 12.** Seed-123 sensitivity campaign on the same 180 SHHS1 participants, extended to E4. The ordering and Holm significance of both locked contrasts are preserved.

| ID        | Subj. macro-F1 | Pooled macro-F1 | Acc.          | $\kappa$      | Contrast | $\Delta$ [CI <sub>95%</sub> ] | $p_{\text{Holm}}$      |
|-----------|----------------|-----------------|---------------|---------------|----------|-------------------------------|------------------------|
| E0        | 0.5303         | 0.5785          | 0.6738        | 0.5420        | E3 – E0  | 0.0331 [0.0225, 0.0438]       | $5.19 \times 10^{-9}$  |
| E2        | 0.5409         | 0.5858          | 0.6874        | 0.5597        | E3 – E6  | 0.0131 [0.0023, 0.0239]       | 0.0085                 |
| E3        | 0.5634         | 0.6031          | 0.7003        | 0.5772        | E4 – E2  | 0.0323 [0.0236, 0.0410]       | $7.40 \times 10^{-13}$ |
| <b>E4</b> | <b>0.5732</b>  | <b>0.6147</b>   | <b>0.7064</b> | <b>0.5869</b> | E3 – E4  | −0.0098 [−0.0125, −0.0073]    | $1.89 \times 10^{-10}$ |
| E6        | 0.5503         | 0.5865          | 0.6873        | 0.5552        |          |                               |                        |

**Table 13.** Locked paired comparisons on SHHS1 subject-mean macro-F1 ( $n = 180$ ). E3 – E2 and the seed-123 E4 rows are post-hoc or secondary and are excluded from the locked family.

| Comparison         | Role           | $\Delta$ | CI <sub>95%</sub>    | $p_{\text{Holm}}$      | W/T/L    | $\Delta_{\text{sign}}$ |
|--------------------|----------------|----------|----------------------|------------------------|----------|------------------------|
| E3 – E0            | locked primary | 0.0412   | [0.0314, 0.0512]     | $2.65 \times 10^{-13}$ | 138/0/42 | +0.533                 |
| E3 – E6            | locked primary | 0.0274   | [0.0182, 0.0370]     | $1.87 \times 10^{-8}$  | 125/0/55 | +0.389                 |
| E1 – E0            | secondary      | 0.00651  | [0.00192, 0.01087]   | 0.00325                | 108/0/72 | +0.200                 |
| E2 – E1            | secondary      | −0.01280 | [−0.02209, −0.00350] | 0.01005                | 80/0/100 | −0.111                 |
| E3 – E2            | post-hoc       | 0.04750  | [0.03724, 0.05792]   | $2.35 \times 10^{-17}$ | 147/0/33 | +0.633                 |
| E4 – E2 (seed 123) | post-hoc       | 0.03232  | [0.02365, 0.04104]   | $7.40 \times 10^{-13}$ | 135/1/44 | +0.508                 |
| E4 – E3 (seed 123) | post-hoc       | 0.00982  | [0.00730, 0.01250]   | $1.89 \times 10^{-10}$ | —        | —                      |

#### 4.6. Which errors should target-domain adaptation prioritise?

E3’s pooled macro-F1 falls from 0.7904 in domain to 0.6099 out of domain, a gap of 0.1805. An aggregate gap does not indicate whether adaptation requires broad retraining or a targeted correction. We therefore localise the loss by class and confusion channel. This is a prioritisation analysis: it identifies where an intervention could recover the most performance, but does not by itself identify the physiological or acquisition mechanism that generated the errors. Throughout this analysis,  $A \rightarrow B$  denotes a reference-labelled A epoch predicted as B; it does not denote a physiological stage transition over time.

The N3 failure is not unique to E3 (Table 15). On the same 22,806 reference N3 epochs, E0 predicts 16,480 as N2 (72.3%) and attains N3 recall 0.2610; E3 predicts 16,674 as N2 (73.1%) and attains recall 0.2582. Their N3 F1 scores are also nearly identical (0.4048 and 0.4055), although E3 has higher pooled macro-F1 overall. The architecture-and-pipeline replacement therefore improves aggregate transfer performance but does not resolve N3 detection.

Four facts constrain the next intervention.

**(i) The degradation is dominated by one class.** N3’s F1 falls by 0.4030; the other four classes fall by 0.09–0.18. N3 alone contributes 45% of the total macro-averaged drop.

**(ii) The model applies a conservative N3 decision in SHHS1.** N3 precision is *higher* out of domain than in (0.944 vs 0.807) while recall collapses (0.258 vs 0.810). The model emits N3 only 0.273× as often as it should, and 16,674 of 22,806 true N3 epochs—73.1%—are labelled N2. This precision–recall asymmetry is consistent with an N3 decision boundary that is too conservative for SHHS1, although it does not reveal whether the cause is montage, age, scoring practice, signal morphology or a combination of these factors.

**(iii) The N3 deficit persists across model families.** E0 and E3 have nearly identical N3 recall and  $N3 \rightarrow N2$  rates despite different epoch encoders, sequence models and input processing. This recurrence argues against attributing the failure specifically to ResNet-1D or TCN; because several components differ simultaneously, it does not identify a common causal mechanism.

**(iv) Label-prior shift is plausible but not sufficient as an explanation.** SHHS1 contains 2.02× the N3 share of Sleep-EDF, while E3 emits N3 at only 0.273× its required rate. It contains 0.38× the N1 share, while E3 over-emits N1 by 2.151×. A prior-shift correction would directionally increase N3 and decrease N1 probabilities [19]; it could therefore mitigate part of this marginal mismatch. We did not fit or validate such a correction, and the concentrated  $N3 \rightarrow N2$  errors near transitions also motivate class-conditional and decision-boundary analyses. The present data neither exclude prior shift nor establish it as the sole cause.

Table 16 makes the E3 prioritisation quantitative. An oracle correction of the single  $N3 \rightarrow N2$  channel in E3 recovers 74.5% of its numerical cross-cohort gap, while the  $N2 \rightarrow \text{REM}$  channel recovers 27.5%. Macro-F1 is nonlinear, so these percentages are not additive; neither value is achievable performance or evidence of a causal mechanism. They rank the two E3 error channels by potential leverage for a subsequent adaptation study. The similar N3 deficit in E0 supports the adaptation target, but the 74.5% value itself is not an E0–E3 architectural contrast.

The second channel,  $N2 \rightarrow \text{REM}$  (11,605 epochs, 15.2% of true N2), shows the mirror-image pattern: REM is over-emitted at 1.477× with recall 0.894 but precision only 0.605. This is a practically relevant limitation of single-channel EEG because EOG and EMG cues are unavailable. E2 is worse still on this channel (REM precision 0.489), so the evaluated preprocessing mitigates but does not remove it.

For completeness, the per-class comparison against E2 confirms that the E3 preprocessing improves every class out of domain, most in REM (+0.0776) and N1 (+0.0618), yet leaves N3 recall almost unchanged (0.2582 vs 0.2496) (Table 18). Thus, the E3 preprocessing bundle does not materially resolve the N3 failure shared by these two fitted pipelines.

Restricting the analysis to the transition neighbourhood tests where a target-domain correction is most needed. Within  $\pm 1$  epoch of a stage change (44,744 epochs, 26.5% of the sample), E3 attains macro-F1

**Table 14.** Per-class transfer of E3 from Sleep-EDF to SHHS1, with the predicted-to-true epoch ratio as a marginal class-emission diagnostic. A ratio of 1.0 means the model emits the class at its true rate; it does not assess probability calibration.

| Stage | Sleep-EDF |           | SHHS1         |               |        |              | $\Delta F1$    |
|-------|-----------|-----------|---------------|---------------|--------|--------------|----------------|
|       | F1        | Pred/True | Prec.         | Rec.          | F1     | Pred/True    |                |
| W     | 0.9304    | 0.993     | 0.8823        | 0.8014        | 0.8399 | 0.908        | −0.0905        |
| N1    | 0.5276    | 1.051     | 0.2528        | 0.5437        | 0.3451 | <b>2.151</b> | −0.1824        |
| N2    | 0.8514    | 0.984     | 0.7394        | 0.7349        | 0.7372 | 0.994        | −0.1143        |
| N3    | 0.8085    | 1.003     | <b>0.9440</b> | <b>0.2582</b> | 0.4055 | <b>0.273</b> | <b>−0.4030</b> |
| REM   | 0.8343    | 1.017     | 0.6052        | 0.8939        | 0.7218 | <b>1.477</b> | −0.1125        |

**Table 15.** Cross-model N3 behaviour on the locked SHHS1 sample. N3→N2 denotes a reference-labelled N3 epoch predicted as N2.

| Configuration              | Pooled macro-F1 | N3 Prec. | N3 Rec. | N3 F1  | N3→N2          |
|----------------------------|-----------------|----------|---------|--------|----------------|
| E0: 15-CNN-BiLSTM          | 0.5761          | 0.9007   | 0.2610  | 0.4048 | 16,480 (72.3%) |
| E3: ResNet-1D-TCN          | 0.6099          | 0.9440   | 0.2582  | 0.4055 | 16,674 (73.1%) |
| E6: E3 + record $z$ -score | 0.5732          | 0.9052   | 0.2005  | 0.3283 | 17,764 (77.9%) |

0.4689, accuracy 0.5290 and  $\kappa = 0.3642$ , against 0.4219, 0.4910 and 0.3188 for E2. N3 recall there falls to **0.0733** while N3 precision remains 0.831: at stage boundaries the model recovers barely one in fourteen true N3 epochs, yet is right five times out of six when it does commit. Of the 7,563 true N3 epochs in this region, 6,855 (90.6%) are labelled N2. Thus, any target-domain intervention should be evaluated explicitly at N2–N3 boundaries rather than only on whole-night macro-F1.

For N1, E3 raises recall by 0.1909 over E2 ( $CI_{95\%}$ : 0.1513–0.2297), the largest per-class preprocessing effect in the study. It does not materially improve N3 over E2.

The E6 sensitivity analysis addresses whether record-wise normalisation is a sufficient generic remedy for the N3 under-emission. It is not: E6 N3 precision, recall and F1 are 0.9052, 0.2005 and 0.3283, respectively, compared with 0.9440, 0.2582 and 0.4055 for E3. Near transitions, E6 N3 recall is 0.0721, essentially unchanged from E3’s 0.0733. E6 emits N3 at only 0.222× its true frequency, compared with 0.273× for E3. This result narrows the adaptation decision: removing record-level amplitude scale, without target labels or a class-specific correction, does not address the dominant transfer error.

**Table 16.** Oracle prioritisation of E3 errors on SHHS1. Each row corrects one confusion channel and recomputes macro-F1. The values quantify potential leverage and are not achievable performance estimates.

| Condition                   | macro-F1 | Gap recovered  |
|-----------------------------|----------|----------------|
| E3 on SHHS1 (observed)      | 0.6099   | —              |
| + N3→N2 errors corrected    | 0.7444   | 0.1345 (74.5%) |
| + N2→REM errors corrected   | 0.6596   | 0.0497 (27.5%) |
| E3 on Sleep-EDF (reference) | 0.7904   | —              |

**Table 17.** Pooled confusion matrix of E3 on SHHS1 (rows: reference; columns: prediction). The two dominant off-diagonal cells are set in bold.

| Reference \ Predicted | W      | N1    | N2            | N3    | REM           |
|-----------------------|--------|-------|---------------|-------|---------------|
| W                     | 29,638 | 4,481 | 1,171         | 21    | 1,672         |
| N1                    | 874    | 3,807 | 601           | 3     | 1,717         |
| N2                    | 2,348  | 5,947 | 56,063        | 324   | <b>11,605</b> |
| N3                    | 78     | 39    | <b>16,674</b> | 5,888 | 127           |
| REM                   | 655    | 784   | 1,311         | 1     | 23,183        |

**Table 18.** Per-class metrics on SHHS1: E3 versus E2 (169,012 scored epochs).

| Stage | E3 Prec. | E3 Rec. | E3 F1  | E2 Prec. | E2 Rec. | E2 F1  | $\Delta F1$ |
|-------|----------|---------|--------|----------|---------|--------|-------------|
| W     | 0.8823   | 0.8014  | 0.8399 | 0.8780   | 0.7275  | 0.7957 | +0.0442     |
| N1    | 0.2528   | 0.5437  | 0.3451 | 0.2367   | 0.3528  | 0.2833 | +0.0618     |
| N2    | 0.7394   | 0.7349  | 0.7372 | 0.7362   | 0.6937  | 0.7143 | +0.0228     |
| N3    | 0.9440   | 0.2582  | 0.4055 | 0.9659   | 0.2496  | 0.3967 | +0.0087     |
| REM   | 0.6052   | 0.8939  | 0.7218 | 0.4886   | 0.9450  | 0.6442 | +0.0776     |

## 5. Discussion

### 5.1. Principal findings

The first research question asked whether the architectural substitutions justified changing the pipeline. They did not establish a stable predictive advantage: the in-domain TCN and ResNet-1D effects were small and non-significant after multiplicity correction, and the ResNet-1D contrast reversed sign on SHHS1. The supported architectural benefit is operational: the implemented ResNet-1D–TCN pipeline reduced measured forward latency and observed training-and-validation wall-clock time, while increasing parameter count and peak GPU memory. It did not establish superior predictive accuracy.

The second question asked which choices retained benefit across cohorts. The fixed-scale E3 procedure outperformed its two locked references on SHHS1, and E3 ranked above E6 under both seeds. Under a fixed ResNet-1D–TCN architecture, the post-hoc E3–E2 contrast was larger than any architectural contrast; the seed-123 E4 extension and the E5 identity audit associate band-pass filtering with most of this difference and provide no evidence that fixed scaling adds a transfer benefit. These secondary contrasts identify preprocessing as the more promising development axis, but they do not establish a causal or universal contribution for any single operation.

The third question asked where adaptation should focus.  $N3 \rightarrow N2$  recurs at essentially the same rate in E0 and E3, while the E3 oracle analysis ranks it above  $N2 \rightarrow \text{REM}$  in potential leverage; the N3 deficit becomes extreme near stage transitions. The evidence therefore identifies a target-domain failure shared across the evaluated pipelines, not an E3-specific weakness. Record-wise  $z$ -scoring does not repair N3, so a labelled, class-specific target-domain calibration experiment is better motivated than another generic architecture or normalisation change.

### 5.2. From error localisation to an adaptation decision

The high precision and low recall of N3, the concentration of N3 errors in N2, and the further recall loss near stage transitions are consistent with a conservative N3 decision boundary in SHHS1. E0’s near-identical N3 recall and  $N3 \rightarrow N2$  rate show that this failure persists across the two model families evaluated; they do not identify a shared causal mechanism. Montage and age are plausible contributors: the model was trained on Fpz–Cz in a younger cohort and evaluated on C4–A1 in an older cohort, while slow-wave amplitude varies with both derivation and age [15, 16]. However, montage, population, acquisition and scoring practice change together between the datasets. The present design cannot assign the failure to one of them, and the precision–recall pattern alone does not identify an amplitude threshold as the mechanism.

The difference in class prevalence is also relevant. A prior-shift correction would move N3 and N1 in the required directions, but it was not fitted or validated here. It should therefore be evaluated as a low-cost baseline in the next target-domain experiment, alongside class-specific calibration and limited fine-tuning; the present error pattern does not justify dismissing it or treating it as a complete solution.

E6 provides an important boundary on the interpretation. If removal of record-level absolute scale were a sufficient correction, E6 should improve N3 recall over E3. Instead, E6 lowers whole-sample N3 recall from 0.2582 to 0.2005 and leaves transition-region recall essentially unchanged (0.0721 versus 0.0733). Thus, the data do not support absolute amplitude scale as a sufficient explanation, and record-wise  $z$ -scoring is not a useful stand-alone adaptation strategy for this failure.

The actionable result is narrower but better supported. Before deployment on SHHS-like data, the next experiment should target the  $N2$ – $N3$  decision boundary using labelled target-domain examples and should report N3 recall specifically near transitions. A small calibration or fine-tuning study on the held-out SHHS adaptation participants would test whether the boundary can be moved without increasing false N3 predictions. The  $N2$ – $\text{REM}$  channel should be the second target. This prioritisation follows directly from the error counts; the choice between montage-aware features, class-specific calibration and target-domain fine-tuning remains an open experimental question.



### 5.3. The roles of the TCN and the ResNet-1D encoder

In-domain, E1 – E0 and E2 – E1 gave positive descriptive gains that did not survive Holm correction. Out of domain, E1 – E0 stayed positive but small while E2 – E1 turned significantly negative. The architectural benefit is therefore domain-dependent, and the claim that ResNet-1D is uniformly superior to the control encoder is not supported—it is contradicted out of domain on unfiltered signals. The dependable advantage of the implemented ResNet-1D–TCN pipeline is operational: competitive accuracy with  $3.76\times$  lower benchmark latency and  $10.2\text{--}12.2\times$  lower observed training-and-validation wall-clock time for the complete ResNet configurations, paid for with  $4.37\times$  the parameters and 28.4% more peak memory. The small E1–E0 latency difference indicates that most of the measured gain is associated with replacing the 15-branch encoder rather than the TCN substitution alone.

### 5.4. Preprocessing as the stronger development axis

The largest observed effect in the study is the post-hoc E3 – E2 contrast on SHHS1 ( $+0.0475$ ,  $\Delta_{\text{sign}} = +0.633$ ), in which architecture is held fixed and only preprocessing changes. It is roughly seven times the largest in-domain architectural contrast. In the seed-123 extension, band-pass-only E4 exceeds E2 and E3; the E5 audit also shows that clipping was inactive in all Sleep-EDF records. These results associate the band-pass condition with most of the observed preprocessing effect and provide no evidence that fixed division by 100 improves transfer. The largest architectural effect measured out of domain was negative. Taken together, these results make preprocessing the stronger development axis in this experiment, while the post-hoc status, one opened external cohort and limited training seeds prevent a universal or causal claim about any individual operation.

### 5.5. Positioning relative to published results

We do not claim state-of-the-art benchmark performance. The wake-trimming policy alone (Section 3.1) moves the wake fraction, and hence accuracy and macro-F1, by an amount comparable to every architectural effect measured here; subject-count, split scheme, and channel choice compound this. Published Sleep-EDF figures for the models discussed in Section 2 [5, 6, 7, 2] were obtained under protocols that differ in several of these dimensions simultaneously, so a numerical comparison would misrepresent the evidence in either direction. Our E0 control is a re-implementation of the ZleepAnlystNet design evaluated under *our* protocol, and it is the reference for the component questions posed here. The study should therefore be read as evidence about relative design choices and transfer failure under a shared protocol, not as a leaderboard claim.

### 5.6. Limitations and a prioritised agenda

Results should be read within the data and protocol evaluated. SHHS1 is one specific out-of-domain sample, not a population. The second seed corroborates direction but two fixed seeds cannot characterise initialisation variance; a proper treatment would require enough seeds to estimate a distribution, and we report no such claim. Several component analyses were run after the sample was opened and are supporting evidence only. Single-channel EEG limits separability for N1 and REM in principle, and the evaluation window uses ground-truth anchors, so it is an offline benchmarking device rather than a deployable rule. We also report no probability-calibration or prior-shift correction analysis: the sequence models are trained with unweighted cross-entropy and their probability estimates are untested, which matters for any clinical use where an abstain option would be valuable [20, 7].

Two reporting gaps remain. Absolute-performance intervals are available for the SHHS configurations (Table 11) but not for the Sleep-EDF ones, where only difference intervals were retained. And although paired win/tie/loss counts and median subject-level differences were preserved for every comparison, the full per-subject macro-F1 distributions were not archived, so we cannot report the dispersion (IQR, range) of individual performance or identify which participants the models handle worst. For a metric whose clinical relevance is inherently per-patient this is a limitation. Per-class SHHS metrics have now been recovered for E6 from the locked prediction artifacts. Pooled class-wise results for E0 have also been recovered from the locked run manifest and support the cross-model N3 comparison; corresponding class-wise results for E1 and transition-region N3 metrics for E0 remain unavailable.

Ordered by expected practical value, we would prioritise: (1) class-specific calibration or limited fine-tuning on the held-out SHHS adaptation participants, evaluated on N3 recall and false-positive rate near N2–N3 transitions; (2) calibration and selective-prediction analysis for all stages; and (3) estimating initialisation variance with a sufficient number of seeds. The first two address the deployment failure identified here; the third improves the strength of the methodological evidence.

## 6. Conclusion

A shared subject-wise protocol allowed six active configurations to be compared on the same Sleep-EDF folds; a seventh planned configuration, E5, was excluded before performance analysis because it was bitwise identical to E4. The TCN and ResNet-1D substitutions did not establish a stable predictive advantage after correction for multiple comparisons. ResNet-1D–TCN nevertheless offers an operational alternative that is  $3.76\times$  faster in the controlled forward benchmark and  $10.2\text{--}12.2\times$  faster in observed training-and-validation wall-clock time, at the cost of more parameters and memory.

On SHHS1, the fixed-scale E3 procedure outperformed both locked references, while the largest observed component difference was associated with preprocessing under a fixed architecture. This result shifts the practical priority from further architecture replacement to signal handling and target-domain evaluation, without establishing any individual preprocessing operation as causal.

Most importantly, the aggregate transfer loss resolves into concrete adaptation targets. E0 and E3 both systematically under-call N3 as N2, showing that E3’s overall transfer advantage does not solve N3 detection; N2 is also frequently over-called as REM by E3. Per-record  $z$ -scoring does not repair the N3 deficit, so neither the evaluated architecture changes nor generic normalisation is sufficient deployment remedy. The evidence supports a targeted, labelled adaptation study on these decision boundaries; it does not yet support an architecture-specific, physiological or clinical claim.

### Reproducibility and data availability

Source code, experimental protocols, data partitions, run manifests and aggregate analysis outputs are retained in the project repository. Sleep-EDF Expanded is publicly available from PhysioNet. SHHS is accessed through the NSRR under its data use agreement; participant-level derived data are not redistributed. Checkpoints are verified by SHA-256 before loading, and the locked SHHS predictions used for the reported analyses were hash-verified against their run manifests. The E6 class-wise results are therefore reported as a descriptive reanalysis of these locked predictions. Protocol-level provenance has not yet been fully reconciled because the protocol file in the current source checkout does not match the protocol hash recorded in the run manifest; consequently, we do not describe this analysis as a rerun from the current protocol. The corresponding protocol version, or a hash-verified reconciliation record, is required before release of the complete reproducibility package.

### Ethics

This study is a secondary analysis of de-identified data from research repositories with their own governance mechanisms; no new participants were recruited. Use of SHHS follows NSRR access conditions. All interpretations apply to the samples and protocols described and do not constitute clinical validation.

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### Declaration of competing interests

The authors declare no competing interests.

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